

Research Project “An Analysis and Simulation of Defending Tracking Control Neural Network for Known Affine Discrete-Time Nonlinear Systems Against Adversarial Attacks” Overview

Overview of existing methods of defending tracking control against adversarial attacks will be performed. Based on obtained results the most efficient approaches of the defense will be selected. The selected methods will be applied to defending tracking control neural network (NN) for known affine discrete-time nonlinear systems against adversarial attacks. For this purpose, the tracking control NN will be described by the difference equation with variable structure. The existence and uniqueness of the correct steady states of the state variable trajectories of the difference equation in the presence of adversarial attacks will be analyzed and shown that they exist and are unique. The stability and convergence to the steady states of the state variable trajectories of the difference equation in the presence of adversarial attacks will be investigated and shown that the trajectories are globally stable and globally convergent to the correct steady states in a finite number of steps. A functional block-diagram of the tracking control NN and its defending part will be presented and analyzed. The diagram will be implemented in MATLAB and/or Python and/or C++ and/or Compute Unified Device Architecture (CUDA). These implementations will be compared with each other. The functional block-diagram will be compared with other competitive analogs and shown that it has less computational and hardware implementation complexity. Computer simulations confirming theoretical derivations and illustrating operation of the defended tracking control NN for known affine discrete-time nonlinear systems will be fulfilled.